

PRIME-LOCAL FABEL FIBRES AND SAME-PRIME NEGATIVE-SQUARE CHANNEL TRANSFERS

THE CLANKERS

ABSTRACT. The reciprocal cubic of every marked first-half Erdős–Straus state is a literal target of the four-dimensional polynomial used in the ES–Fabel and Split-Zero conductor comparison. We determine its complete local fibre algebra at the original prime. Exterior states yield two ramified quadratic factors; middle states yield two unramified quadratic factors. Both have exactly three rational local inverse points, with all seven geometric affine branches and integral specializations accounted for. Separately, negative-four-square shape coordinates give original same-prime middle states on an exact divisor domain, covering entire unbounded shape lines. All arithmetic coordinates, fibres and ordered denominator returns are explicit. Neither conclusion proves universal Erdős–Straus occupancy.

1. RESULT AND SCOPE

This is a continuation of the unresolved Turn 7 existence task. The original four-dimensional Fabel polynomial in the Zeta workbench admits a literal arithmetic specialization at the reciprocal cubic of an Erdős–Straus state. Its two channels have different local fibre algebras at the original prime: the exterior channel supplies ramified quadratic branches, whereas the middle channel supplies unramified quadratic branches. Both have exactly three rational local inverse points. The derivative, the full seven-point affine fibre, its integral specialization and the missing projective sign are retained.

A separate constructive continuation extends the earlier bounded-shape E-to-M returns to arbitrary negative-four-square shape coordinates on their exact integer domain. In particular the entire lines $\alpha = -4$ and $\beta = -4$ have original middle returns. Neither result produces the initial state at an arbitrary prime. No universal ES assertion, completed Turn 7 existence milestone, historical-priority determination or whole-programme analytic identification is claimed.

The polynomial and global inverse formulas are antecedents from the Zeta source [2, GF1–11, FC32–35]. Its four-dimensional map is the ES reader’s extension, not an additional preimage attributed to the original three-dimensional Alpöge–Fabel example discussed by Tao [6]. The classical Type-I/II coordinates are credited to [1, §2].

2. ORIGINAL ARITHMETIC AND THE COMMON RECIPROCAL CUBIC

Fix a prime $p \equiv 1 \pmod{4}$. Retain an original first-half source

$$p/4 < a < p/2, \quad R = 4a - p, \quad u \mid a^2, \quad E : R \mid 4u + 1, \quad M : R \mid u + a.$$

For $a = \prod_l l^{e_l}$ retain $\beta_l = v_l(u) - e_l \in [-e_l, e_l]$. With $g = \gcd(a, u)$ the original normalization is

$$h = g^2/u, \quad r = u/g, \quad s = a/g, \quad a = hrs, \quad u = hr^2, \quad \gcd(r, s) = 1.$$

For E let $\kappa = (pr + s)/R$ and return $(x_1, x_2, x_3) = (a, h s \kappa, p h r \kappa)$. For M let $\lambda = (r + s)/R$ and return $(x_1, x_2, x_3) = (a, p h s \lambda, p h r \lambda)$. Multiplication verifies the reciprocal identity. The ordered inverses are $u = a^2/(R x_2 - p a)$ and $u = p a^2/(R x_2 - p a)$, respectively. The M complement $u \mapsto a^2/u$ exchanges r, s and the last two denominators; its orientation is retained rather than suppressed.

Every normalization factor is a p -unit. For E use $pD + 1 = 4hr\kappa$, with $D = (4u + 1)/R = (r + \kappa)/s$; in particular κ need not be smaller than p . For M, $h, r, s < p$ and $\lambda = (r + s)/R < p$. The three denominators are distinct. In E, $\kappa > r$ and $pr > s$; in M, $r = s$ would force $r = s = 1$ and $R \mid 2$. Their first denominator is below $p/2$ and the other two are above $p/2$.

Lemma 2.1 (Original character distinction). *For every original divisor $d \mid a^2$, $(d/R) = (d/p)$, with Jacobi symbol on the left. Every E state has $(h/p) = -1$. Every M state has $(rs/p) = -1$.*

Proof. For odd $l \mid a$, quadratic reciprocity and $R \equiv -p \pmod{l}$ give $(l/R) = (-1/l)(R/l) = (p/l) = (l/p)$. If $2 \mid a$, then $R \equiv -p \pmod{8}$ and the supplementary law gives $(2/R) = (2/p)$. Multiply over the actual exponents. At E, $u \equiv -1/4 \pmod{R}$ and $R \equiv 3 \pmod{4}$, so $(u/p) = -1$. Since $u = hr^2$, this proves the first assertion. At M, $r \equiv -s \pmod{R}$ and all the factors are units, so $(r/p) = (r/R) = -(s/R) = -(s/p)$. This proves $(rs/p) = -1$. These are the classical laws in [4, §6.4], applied to the retained original factors. $\square$

Define the labelled reciprocal roots and polynomial

$$t_i = p/x_i, \quad f(T) = \prod_{i=1}^3 (T - t_i) = T^3 - 4T^2 + \sigma_2 T - \sigma_3. \quad (1)$$

Then $2 < t_1 < 4$ and $0 < t_2, t_3 < 2$. The target for the *existing* Fabel map is

$$\mathbf{U} = (0, -4, \sigma_2, -\sigma_3), \quad H_{\mathbf{U}}(U, V) = V \prod_i (U - t_i V). \quad (2)$$

The leading zero and the coefficient of $U^3 V$, namely one, are fixed, not rescaled away. The inverse on the actual arithmetic domain is $x_i = p/t_i$, with its positivity and integrality tests and the original E/M divisor inverse. The coefficient target alone forgets the ordering; p and the labels are kept.

The preceding shape curve has different coordinates: $\alpha = v - R, \beta = v - D, X = v, Y = 2a$, with the additional fixed-prime line $2Y = p + X - \alpha$. Its inverse is $a = Y/2, R = X - \alpha, D = X - \beta, u = (RD - 1)/4$. Thus the connection to (2) is the explicit composition through this original state. No identification of two unmarked cubic curves is asserted.

3. THE LITERAL FABEL MAP AND EVERY INVERSE BRANCH

Use ξ for its first source coordinate, distinct from the original a . Fix $i^2 = -1$ and put $P = (P_0, P_2, P_3, P_4)$, where

$$\begin{aligned} P_0 &= \xi^3 z + 2\xi^2 y - i\xi, \\ P_2 &= -\xi^3 y^2 z - 2i\xi^2 yz + \xi^2 w - 2\xi^2 y^3 - 10i\xi y^2 + 3\xi z + y, \\ P_3 &= 2\xi^3 y^3 z + 6i\xi^2 y^2 z + 2\xi^2 wy + 4\xi^2 y^4 + 2i\xi w - 4i\xi y^3 - 2\xi yz + 2iz + 7y^2, \\ P_4 &= 2\xi^3 y^4 z + 8i\xi^2 y^3 z + \xi^2 wy^2 + 4\xi^2 y^5 + 2i\xi wy + 7i\xi y^4 - 10\xi y^2 z - 4iyz - w - 3y^3. \end{aligned} \quad (3)$$

Its factor coordinates are

$$\begin{aligned} b &= i + \xi y, & c &= -i + 2\xi y + \xi^2 z, \\ d &= -iy - \xi(iz + 2y^2) - \xi^2 yz, & e &= 2z - 7iy^2 + \xi w, \\ f_0 &= iw + 3iy^3 - 4yz + \xi(6iy^2 z + wy + 4y^4) + 2\xi^2 y^3 z. \end{aligned}$$

Writing $L = \xi U + bV$ and $C = cU^3 + dU^2V + eUV^2 + f_0V^3$, substitution gives

$$\xi d + bc = 1, \quad C(-b, \xi) = 1, \quad LC = P_0 U^4 + U^3 V + P_2 U^2 V^2 + P_3 UV^3 + P_4 V^4. \quad (4)$$

For a simple finite root t_i of (1), let $d_i = f'(t_i) = (t_i - t_j)(t_i - t_k)$. The full inverse is

$$\boxed{\xi^2 = d_i^{-1}, \quad y = -t_i - i/\xi, \quad z = 2t_i/\xi + 3i/\xi^2, \quad w = 7it_i^2/\xi + (-4 - 17t_i)/\xi^2 - 13i/\xi^3.} \quad (5)$$

Both signs of ξ are retained. The finite marked root is recovered by $t_i = -b/\xi$, and $C(-b, \xi) = \xi^2 f'(t_i) = 1$. Coefficient comparison in (4) determines the quotient factor uniquely; the factor-coordinate inverse $y = (b - i)/\xi, z = (c + i - 2\xi y)/\xi^2, w = (e - 2z + 7iy^2)/\xi$ then proves (5) and its exhaustiveness.

At $\xi = 0$ the single chart inverse is

$$q_\infty = (0, -4, i(112 - \sigma_2)/2, -704 + 8\sigma_2 + \sigma_3). \quad (6)$$

It has $b = i, c = -i$. The other signed projective infinity factor has $b = -i$ and is outside this affine chart. Thus (2) has *seven*, not eight, geometric affine preimages: six finite ones and (6). The missing infinity sign is different from any arithmetic field or integrality obstruction.

For completeness the constant Jacobian statement used here is verified as follows. On $\xi \neq 0$ use (ξ, t, A, B) , with $A = P_0, B = P_2$. The target equations are

$$P_3 = \xi^{-2} - 4At^3 - 3t^2 - 2Bt, \quad P_4 = 3At^4 + 2t^3 + Bt^2 - t\xi^{-2}.$$

The full source inverse has determinant $-\xi^{-5}$ in those coordinates; the target determinant is $2\xi^{-5}$. Their ratio is -2 . Polynomial continuation proves $\det DP = -2$ everywhere. These are the antecedent GF10–11 calculations [2], with their constant and sign retained; the verifier independently expands the original derivative at every recorded inverse point.

4. THE COMPLETE LOCAL ALGEBRA DISTINGUISHES E FROM M

Fix either root $i_p \in \mathbb{Q}_p$ of -1 . Its existence follows by lifting a root modulo $p \equiv 1 \pmod{4}$. Normalize $v_p(p) = 1$ on all extensions. The reduced coordinate algebra of the actual fibre is

$$\mathcal{A}_{p,x} = \mathbb{Q}_p \times \prod_{i=1}^3 \mathbb{Q}_p[\xi_i]/(\xi_i^2 - d_i^{-1}). \quad (7)$$

The first factor is the single infinity chart. Formula (5) and its inverse identify the other factors; the derivative is nonzero, so there are no unrecorded nilpotents. This is the coordinate algebra of the fixed target, not a newly chosen seven-point model.

Theorem 4.1 (Prime-local arithmetic fibre). *For every original first-half ES state at a prime $p \equiv 1 \pmod{4}$, $P^{-1}(\mathbf{U})$ has exactly three $\mathbb{Q}_p$ -rational points. More precisely,*

$$\begin{aligned} E : \quad \mathcal{A}_{p,x} &\simeq \mathbb{Q}_p^3 \times K_{\text{ram}} \times K_{\text{ram}}, \\ M : \quad \mathcal{A}_{p,x} &\simeq \mathbb{Q}_p^3 \times K_{\text{unr}} \times K_{\text{unr}}, \end{aligned} \quad (8)$$

where $K_{\text{ram}}/\mathbb{Q}_p$ is a ramified quadratic extension and $K_{\text{unr}}/\mathbb{Q}_p$ is the unramified quadratic extension. The repeated field factors retain their distinct root labels. There is no fully split seven-point local fibre in the original arithmetic domain.

Proof. For E put $H = hrs\kappa$. Then

$$\begin{aligned} (t_1, t_2, t_3) &= H^{-1}(p\kappa, pr, s), \\ (d_1, d_2, d_3) &= H^{-2}(p(\kappa - r)(p\kappa - s), -p(\kappa - r)(pr - s), (p\kappa - s)(pr - s)). \end{aligned} \quad (9)$$

All factors h, r, s, κ are p -units. Also $p \nmid \kappa - r$: otherwise $4hr\kappa \equiv 1 \pmod{p}$ would make $4hr^2 \equiv 1$, contrary to Lemma 2.1. Hence

$$(v_p(d_1), v_p(d_2), v_p(d_3)) = (1, 1, 0), \quad d_1/d_2 \equiv -1 \pmod{p}, \quad d_3 \equiv 16 \pmod{p}. \quad (10)$$

A unit congruent to a nonzero square modulo the odd prime p is a square in $\mathbb{Q}_p$ by Hensel lifting. Thus d_1/d_2 is a square and the first two quadratic factors are copies of one ramified field (their defining square classes have odd valuation). The last quadratic splits into two rational points. Together with infinity this proves the E case.

For M put $H = hrs\lambda$. Then

$$(t_1, t_2, t_3) = H^{-1}(p\lambda, r, s).$$

The numerator differences are units: $r, s < p$, $r \neq s$, and $p \nmid r - s$. Modulo p , the derivative classes, up to the common square H^{-2} , are

$$rs, \quad r(r - s), \quad s(s - r). \quad (11)$$

The first is a nonsquare by Lemma 2.1. Their product is $-r^2s^2(r - s)^2$, a square since -1 is a square modulo p . Thus exactly one of the three classes is a square, and exactly two are nonsquares. The two nonsquare units have square ratio and define the same unramified quadratic extension. The remaining quadratic gives two rational points. This proves (8) and the rational-point count. $\square$

The local facts used in this proof can be made constructive. A root b_n of a unit z modulo p^n has the unique lift

$$b_{n+1} = b_n + p^n t, \quad t \equiv -\frac{b_n^2 - z}{p^n} (2b_n)^{-1} \pmod{p}.$$

Use the compatible rational residue of z to compute the integer quotient. A nonsquare residue has irreducible separable quadratic reduction, giving the unramified degree-two extension; a class p times a unit has square root of valuation $1/2$, giving ramification index two. These are the standard local field facts in [5]; the displayed lift is also checked directly.

An algebra isomorphism over $\mathbb{Q}_p$ must permute the primitive field factors. It cannot change their ramification indices. Therefore no such isomorphism carries the E fibre algebra to the M fibre algebra at the fixed prime. Equality of the rational-point counts (both three) does not give that isomorphism. This restricts an *unchanged fibre algebra*; it does not prohibit an arithmetic map to a different target, constructed below.

5. INTEGRAL SPECIALIZATION AND THE NONPROPER BOUNDARY

For E, the exact numerator formula further gives

$$v_p(\sigma_2) = 1, \quad v_p(\sigma_3) = 2, \quad v_p(\text{Disc } f) = 2, \quad \bar{f}(T) = T^2(T - 4). \quad (12)$$

Indeed $\sigma_2 = p\{p\kappa r + s(\kappa + r)\}/H^2$ and $\kappa + r = sD$ is a unit. If $p \mid D$, then $RD = 4u + 1$ would make the nonresidue u equal to the square $-1/4$ modulo p , impossible. The product gives σ_3 and the three differences give the discriminant.

For each of the four signed branches over t_1, t_2 , equation (5) has the exact valuation vector

$$(v_p(\xi), v_p(y), v_p(z), v_p(w)) = (-1/2, 1/2, 1, 1). \quad (13)$$

The least terms are respectively ξ , $-i/\xi$, $3i/\xi^2$ and $-4/\xi^2$. The coefficient 3 is a unit since $p \equiv 1 \pmod{4}$. There is no equal-valuation cancellation in these minima. These are genuine generic-field points but are not integral and have no affine specialization over the valuation ring. The two branches at t_3 and the infinity branch are integral.

Over an algebraic closure of the residue field, the complete E special fibre has precisely the following three points:

$$(0, -4, 56i_p, -704), \quad (1/4, -4 - 4i_p, 32 + 48i_p, -1152 - 384i_p), \\ (-1/4, -4 + 4i_p, -32 + 48i_p, -1152 + 384i_p). \quad (14)$$

The double root zero has no resultant-one factor scaling; the simple root four has derivative sixteen. Thus the reduction from seven geometric generic points to three special points is a loss at the affine boundary, not a vanishing source Jacobian. The determinant -2 remains a unit.

For M, all three roots and all derivative values are units or zero with distinct reductions: the roots are $0, r/H, s/H$. Every geometric inverse in (5) is integral over an unramified extension. Its geometric special fibre still has seven points; precisely three are rational over $\mathbb{F}_p$. Thus counting only rational special points also fails to distinguish the two channels.

The exact relation to the original conductor. The Zeta source constructs fixed invertible complex linear maps Φ, Ψ with $T_A \Phi = \Psi$ and transported polynomials $\Phi P \Phi^{-1}, \Psi P \Psi^{-1}$ [2, FC32–35]. Consequently (5)–(6) transport point by point through that exact complex map, with its retained original Gram G and norm $q^* G q$. No Euclidean replacement, new coefficient fit or linear-conductor kernel is asserted. The p -adic statements here concern the original polynomial over $\mathbb{Q}(i)$ after the specified embedding in $\mathbb{Q}_p$. Complex period entries in Ψ are not thereby declared to have a $\mathbb{Q}_p$ -model. Such a descent would be an additional theorem.

A nonempty polynomial fibre is not an arithmetic solution. The target $(0, -4, 0, 1)$ has the exact infinity inverse $(0, -4, 56i, -705)$. Its cubic $T^3 - 4T^2 + 1$ has no rational root: the rational-root theorem leaves only ± 1 , neither a root. Thus it cannot be the reciprocal cubic of three positive integer denominators at any rational p . The inverse formula alone does not supply the initial arithmetic target.

6. THE EXACT INTEGRAL ROOT ORDER AND ITS OBLIGATORY DENOMINATOR

This is a further arithmetic distinction between the coefficient target and its split roots. Let $\mathcal{O}_f = \mathbb{Z}_p[T]/(f)$, observed by evaluation at the three original labelled roots. All are integral, and f is monic.

Theorem 6.1 (Root-order defect and exact interpolation loss). *For an original E state,*

$$\mathcal{O}_E \simeq \{(z_1, z_2, z_3) \in \mathbb{Z}_p^3 : z_1 \equiv z_2 \pmod{p}\}. \quad (15a)$$

Its index in $\mathbb{Z}_p^3$ is p and its conductor ideal in that ring is $p\mathbb{Z}_p \times p\mathbb{Z}_p \times \mathbb{Z}_p$. For M, evaluation gives $\mathcal{O}_M \simeq \mathbb{Z}_p^3$.

Given any original E and M state at the same prime, let x_i and y_i be their ordered reciprocal roots. There is a unique polynomial F of degree at most two taking x_i to y_i , and a unique G of degree at most two taking y_i to x_i . Their exact formulas are

$$F(T) = \sum_i y_i \frac{(T - x_j)(T - x_k)}{(x_i - x_j)(x_i - x_k)}, \quad G(T) = \sum_i x_i \frac{(T - y_j)(T - y_k)}{(y_i - y_j)(y_i - y_k)}. \quad (15b)$$

Writing coefficients in increasing degree, F has exact p -valuations $(0, -1, -1)$, while $G \in \mathbb{Z}_p[T]$. Moreover $pF \pmod{p}$ is a nonzero scalar multiple of $T(T-4)$. The compositions satisfy $G(F(T)) = T \pmod{f_E}$ and $F(G(T)) = T \pmod{f_M}$.

Proof. For E, $x_1 - x_2$ has valuation one; both differences from x_3 are units. A polynomial over $\mathbb{Z}_p$ therefore has its first two values congruent modulo p . Conversely, for a tuple with that congruence, interpolation at x_1, x_2 is integral since its divided difference is integral. Its error at x_3 can be corrected by an integral multiple of $(T - x_1)(T - x_2)$, whose value there is a unit. This proves (15a). The quotient is exactly $\mathbb{F}_p$ by $z \mapsto z_1 - z_2 \pmod{p}$. Multiplication by an arbitrary tuple preserves the congruence exactly when its first two coordinates are divisible by p , which proves the conductor. All three differences for M are units, giving its integral interpolation inverse.

Formula (15b) is the original labelled interpolation. The E derivative valuations imply $pF \in \mathbb{Z}_p[T]$. It cannot be integral before that multiplication, since it takes two congruent arguments x_1, x_2 to the incongruent values y_1, y_2 . In fact modulo p , only its second Lagrange summand survives after multiplication by p : y_1 is divisible by p , while the third denominator is a unit. Therefore

$$pF(T) \equiv (py_2/d_2)T(T-4) \pmod{p},$$

with a nonzero coefficient. This proves valuation -1 for the linear and quadratic coefficients. The constant term is a unit: its second summand has valuation zero, and the first and third have positive valuation. The reverse interpolation has only unit denominators and integral numerators. Equality of each composition with T at all three distinct labelled roots proves the two polynomial congruences. $\square$

The exact sequence

$$0 \longrightarrow \mathcal{O}_E \longrightarrow \mathbb{Z}_p^3 \xrightarrow{z \mapsto z_1 - z_2 \pmod{p}} \mathbb{F}_p \longrightarrow 0$$

has no $\mathbb{Z}_p$ -linear right inverse, because the middle module is torsion-free. Tensoring with $\mathbb{Q}_p$ removes this index- p quotient; that operation loses original integral information. This is the root-order conductor, explicitly distinct from the Zeta workbench's weighted linear conductor T_A . We do not identify their source modules.

Thus there is a fully explicit rational root-level map between the two cubics, with a sharp integral obstruction and an integral reverse map. The lack of an isomorphism between their seven-point local fibre algebras is not described as unrelatedness. It is the additional quadratic-factor ramification in (8), which this root interpolation alone does not preserve.

For a small complete illustration, at $p = 5$ take the original E denominators $(2, 4, 20)$ and M denominators $(2, 10, 5)$. Their reciprocal root triples are $(5/2, 5/4, 1/4)$ and $(5/2, 1/2, 1)$. The

exact maps are

$$F(T) = \frac{14}{15}T^2 - \frac{19}{10}T + \frac{17}{12}, \quad G(T) = \frac{7}{4}T^2 - \frac{37}{8}T + \frac{25}{8}.$$

The first requires exactly one power of five in its local denominator; $5F \equiv 3T(T-4) \pmod{5}$. The reverse is integral at five. Both ordered root triples and their original denominator inverses are retained in the certificate.

7. AN UNBOUNDED FAMILY OF EXACT SAME-PRIME CHANNEL RETURNS

Continue with the original E state, and set $\alpha = hs^2 - R$, $\beta = hs^2 - D$. The earlier shape identities give

$$h \mid \alpha\beta - 1, \quad p \equiv \alpha \pmod{h}, \quad p\beta \equiv 1 \pmod{h}. \quad (15)$$

Theorem 7.1 (Negative-four-square channel transfer). *Suppose one marked coordinate is $-4c^2$, with $c > 0$ integral. Then $h \equiv 3 \pmod{4}$ and $\gcd(h, c) = 1$. Put $j = (h+1)/4$. If $c \mid j$, set*

$$Q = h, \quad U = \begin{cases} c^2, & \alpha = -4c^2, \\ (j/c)^2, & \beta = -4c^2, \end{cases} \quad R_M = (p+4U)/Q, \quad a_M = (p+R_M)/4. \quad (16)$$

Then $U \mid j^2$, $Q \mid p+4U$, $0 < R_M < p$, and (16) is an original M state. With $g_M = \gcd(j, U)$, its complete normalization is

$$h_M = g_M^2/U, \quad r_M = U/g_M, \quad \lambda_M = j/g_M, \quad s_M = R_M\lambda_M - r_M. \quad (17)$$

The new h_M is a square. The ordered return is $(a_M, ph_Ms_M\lambda_M, ph_Mr_M\lambda_M)$. Both entire shape lines $\alpha = -4$ and $\beta = -4$ are covered, without a bound on their other coordinate.

Proof. Equation (15) makes h odd and coprime to c . At every odd prime dividing h , either $p \equiv -4c^2$ or $p^{-1} \equiv -4c^2$. Reciprocity gives $(h/p) = (-1/h)$. Lemma 2.1 makes this -1 , so $h \equiv 3 \pmod{4}$. In the α case the required congruence is immediate. In the β case, $p \equiv -(4c^2)^{-1} \pmod{h}$ and $4j \equiv 1 \pmod{h}$, so $p + 4(j/c)^2 \equiv 0$. Availability follows from $c \mid j$ in both cases.

For α , the original inequality $R = hs^2 + 4c^2 < p$ implies $4U < p - h < (h-1)p$. For β , use $4U \leq 4j^2 = (h+1)^2/4$ and $p > 2h$; for $h \geq 3$, $(h+1)^2 < 8h(h-1)$, so again $4U < (h-1)p$. Thus $R_M < p$. Its residue is $3 \pmod{4}$, proving the original first-half range.

The valuation normalization of $U \mid j^2$ gives $h_Mr_M\lambda_M = j$, $h_Mr_M^2 = U$, and $\gcd(r_M, \lambda_M) = 1$. Also $a_M = R_Mj - U = h_Mr_Ms_M > 0$. Since $r_M \leq j < p$, the equation $QR_M = p + 4U$ proves $\gcd(r_M, R_M) = 1$, hence $\gcd(r_M, s_M) = 1$. This is therefore precisely the original M normalization.

For an explicit square-grade formula write $j = cm$, $d = \gcd(c, m)$. In the α case, $(h_M, r_M, \lambda_M) = (d^2, c/d, m/d)$; in the β case it is $(d^2, m/d, c/d)$. This also proves that the original shared budget has not been used twice. $\square$

The condition $c \mid j$ is the exact availability domain of the two selected cofactor words: $c^2 \mid j^2$ if and only if $c \mid j$, and j/c must be integral. It is not a hypothesis inserted into a universal existence theorem. Its failure can leave other words available, or the entire selected cofactor box empty. The square-divisor criterion in (16) is classical Type-II arithmetic; what is proved here is its supply from these actual exterior shapes.

7.1. Complete inverse fibres and coefficient collisions. The target retains its cofactor $Q = 4h_Mr_M\lambda_M - 1$, so it recovers the original h . In the α branch it recovers $c = \sqrt{U}$; in the β branch $c = j/\sqrt{U}$. The square roots here are positive integers.

For fixed (p, h, c, α) , put $N = (p + 4c^2)/h$. Every original preimage is obtained by

$$s \mid N, \quad k = N/s, \quad s + k \equiv 0 \pmod{4}, \quad r = (s + k)/4, \quad a = hrs, \quad u = hr^2,$$

with the original coprimality and first-half inequalities. Conversely $4rs - s^2 = N$ follows directly from the shape, so this is the full fibre.

For fixed (p, h, c, β) , put $F = (4c^2p + 1)/h$. For $1 \leq s \leq \lfloor (p-1)/(2h) \rfloor$, set

$$D = hs^2 + 4c^2, \quad \Delta_s = s^2(D^2 - p) - F.$$

Keep exactly the positive squares $k^2 = \Delta_s$ for which $sD - k$ is positive and even, and set $r = (sD - k)/2$, $\kappa = sD - r$. The original coprimality and range tests complete the fibre. This comes from $4hr(sD - r) = pD + 1$ and $k = \kappa - r > 0$, so both directions are proved.

Within each marked branch, c is recovered. Across an α branch with c_1 and a β branch with c_2 , the selected target divisor is the same exactly when $c_1c_2 = j$. Different original sources in any one fibre remain separate integer coefficients. Pushing them to one target sums those coefficients; the kernel is generated by their same-target differences. No injectivity or one-to-one state count is claimed after those records are discarded.

7.2. An actual transition, with local ramification changed. At the hard prime $p = 5209$, the original E state is

$$(a, R, u, h, r, s, \kappa) = (1330, 111, 18620, 95, 14, 1, 657).$$

Its shape is $(-16, -576)$, so the two selected branches have $c_1 = 2, c_2 = 12$ and $j = 24$. Both supply the *same* target $U = 4$. Formula (17) gives

$$(a_M, R_M, u_M, h_M, r_M, s_M, \lambda_M) = (1316, 55, 4, 4, 1, 329, 6),$$

$$\boxed{\frac{4}{5209} = \frac{1}{1316} + \frac{1}{41130264} + \frac{1}{125016}}. \quad (18)$$

The original E box uses primes 2, 5, 7, 19. Its M target uses 2, 7, 47; the factor 47 is part of the reconstructed integer, not an available original E factor silently adjoined. At this same prime the original Fabel fibre changes from ramified to unramified by Theorem 4.1, because the target cubic has changed through an actual arithmetic return. No local algebra isomorphism between the unchanged fibres is being asserted.

A different hard example is

$$p = 4561, \quad (a, R, u, h, r, s, \kappa) = (1593, 1811, 43011, 59, 27, 1, 68), \quad \beta = -36.$$

Here $c = 3, j = 15$ returns

$$(a_M, R_M, u_M, h_M, r_M, s_M, \lambda_M) = (1160, 79, 25, 1, 5, 232, 3),$$

with denominators (1160, 3174456, 68415).

At the hard prime $p = 3049$, the original E state $(a, R, u, h, r, s, \kappa) = (806, 175, 20956, 31, 26, 1, 453)$ has $\alpha = -144$, hence $c = 6, j = 8$. The selected $U = 36$ is unavailable in $j^2 = 64$. More strongly, the entire cofactor box has divisors 1, 2, 4, 8, 16, 32, 64, none satisfying $31 \mid 3049 + 4U$ (the required residue is 5 (mod 31)). A different original shell has the middle state $(a, R, u, h, r, s, \lambda) = (770, 31, 5, 5, 1, 154, 5)$. This is an exact local obstruction, not an ES counterexample.

8. VERIFICATION, SOURCE DISTINCTIONS AND REMAINING TASK

The release verifier enumerates the complete first-half boxes for all primes $p \equiv 1 \pmod{4}$ through its declared bound, checks every labelled E/M state, its ordered identity, the local field signatures, and every present negative-square crossing. It evaluates (3) on all seven exact branches of representative fibres in the four-dimensional rational algebra generated by i and ξ ; it does not assume that this algebra is a field. It computes the original Jacobian by exact first-order arithmetic and checks the unit-root lifts modulo p^4 .

The separate program imports none of the main implementation or predecessor software. It enumerates primitive h, r, s to recover the full state lists, checks the factor-coordinate product (4), independently enumerates the inverse crossing fibres, and exhausts the original affine polynomial fibre over $\mathbb{F}_{17}$ at both special targets. Execution receipts distinguish these checks from a mathematical review or a Lean build.

The positive progress is an explicit same-prime return on unbounded shape lines, and an exact local arithmetic meaning for the inverse branches read in the Zeta workbench. The root-field signature does not prove that an ES reciprocal target exists, nor that every remaining exterior

shape reaches a middle state. Turn 7’s universal positive-coefficient obligation remains unresolved. We do not advance to Turn 8 under an assumed occupancy statement.

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